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Souvik Sarkar

GitHub: SouvikSarkarJr
Portfolio: ssjr/portfolio

Education :

Master of Technology in Computer Science and Engineering **2026-2028**
Techno Main Saltlake (TMSL)

Obtained CGPA: Expected 2028

Bachelor of Technology in Computer Science and Engineering **2022-2026**
Techno India University (TIU)

Obtained CGPA: 8.19 out of 10

Publication :

Self-Adaptive Improved Fungal Growth Optimization: A Hybrid Metaheuristic Framework for High-Quality, Robust and Scalable Clustering *Knowledge and Information Systems (Springer Nature)*
October 2026 [View Paper]

Projects :

Bi-Directional Sign Language Translator [View Project] **September 2026**
Tech Stack: Python, PyTorch, MediaPipe & Hugging Face.

- Built a unified dual-path translation framework enabling real-time, mutual conversion between continuous sign gestures and written English.
- Extracted 3D keypoints via MediaPipe, **reducing raw input dimensional overhead by over 70%** while retaining motion dynamics.
- Fine-tuned a **seq2seq Transformer model** in PyTorch, enabling **real-time inference at 30+ FPS** with high BLEU scores.

FGOIKMeans Hybrid Optimization [View Project] **August 2025**
Tech Stack: Python, NumPy, Pandas, Scikit-learn & Matplotlib.

- Designed and implemented a hybrid machine learning algorithm pairing Fungal Growth Optimization (FGO) with K-Means to resolve local optima stagnation.
- Achieved an **18.92% increase** in clustering quality across standard multi-dimensional benchmark datasets.
- Accelerated **convergence rates** and enhanced **model stability** by eliminating K-Means sensitivity to random centroid initialization.

Skills :

Languages	Python, C, C++
AI/ML Frameworks	PyTorch, Scikit-learn, Hugging Face, SciPy, scikit-fuzzy.
Data & Libraries	Pandas, NumPy, Matplotlib, GeoPandas, Pygame.
Web Technologies	HTML, CSS, JavaScript.
Developer Tools	GitHub, VS Code, LaTeX.
Documentation	Google Workspace, Microsoft Office.